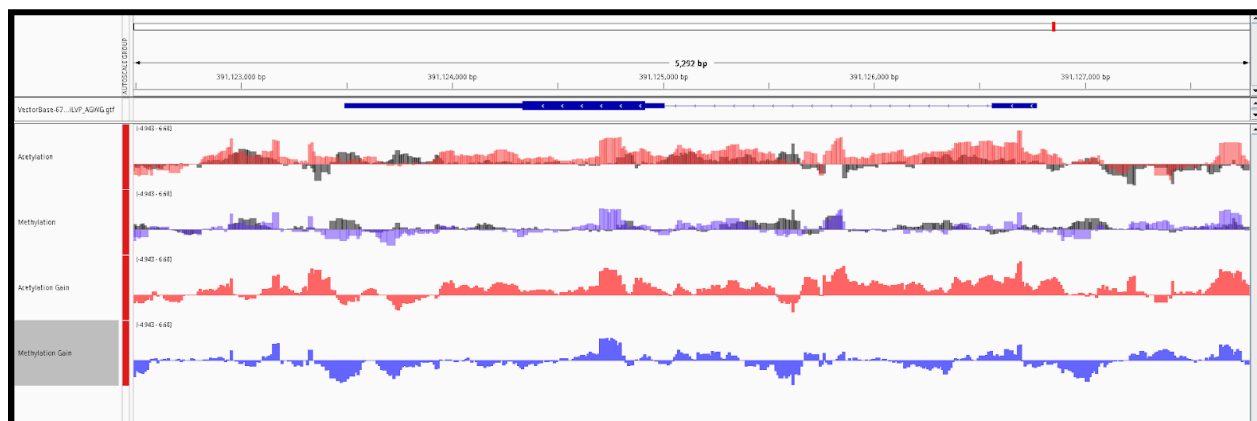


Using IGV to Visualize ChIP-Seq Data

Last updated Feb 11, 2025 by Zoey Mikol

IGV is a tool for exploring/visualizing different kinds of genomic data. In this document, we will mainly focus on creating visualizations of ChIP-Seq read density:



Don't be intimidated by the length of this document; it is simply due to the many images.

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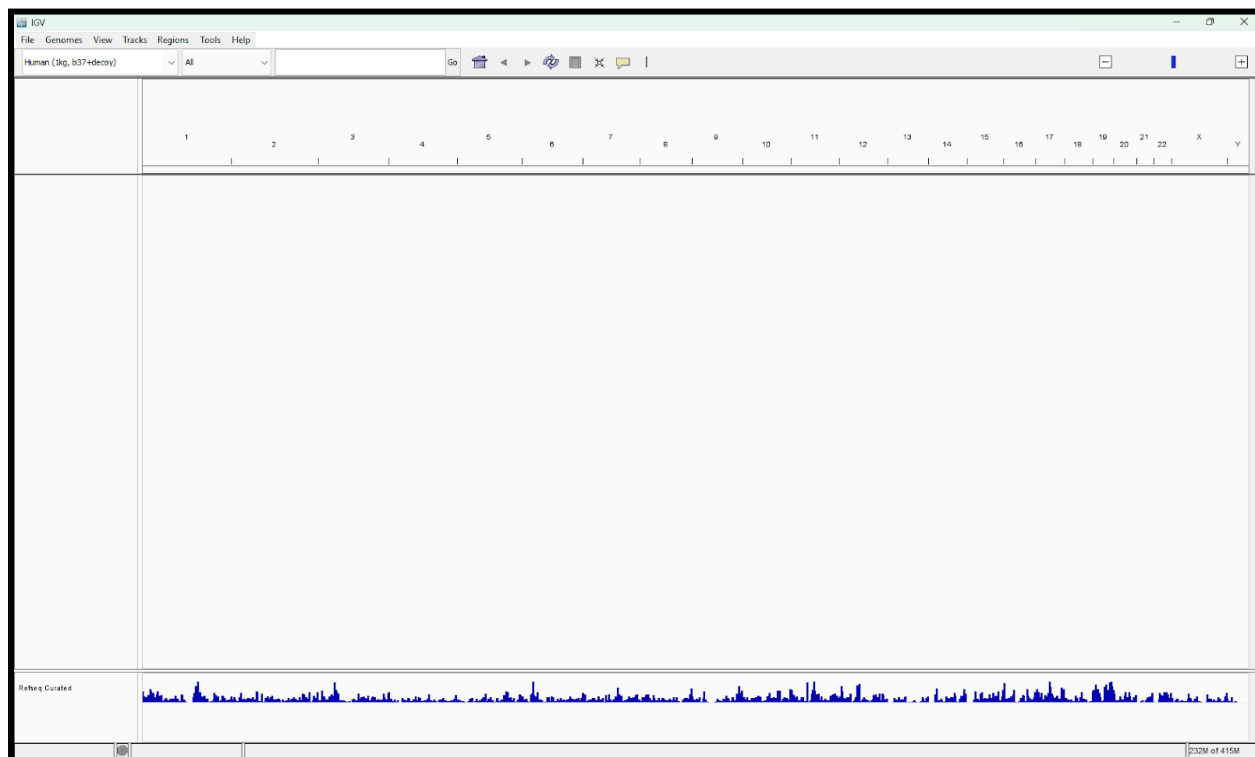
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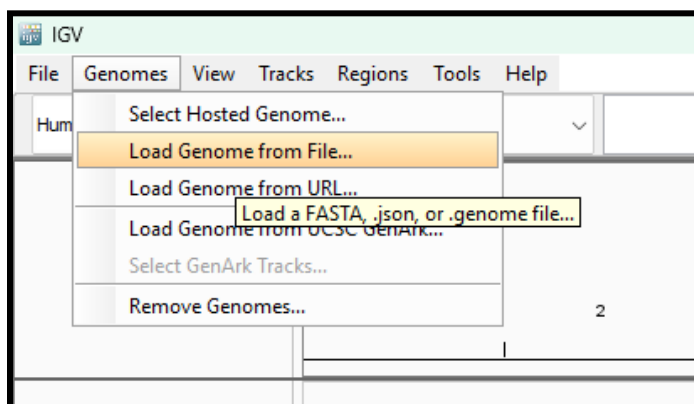
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7. Saving
 - a. Saving visualization images
 - b. Saving your IGV session (**READ THIS BEFORE CLOSING IGV**)

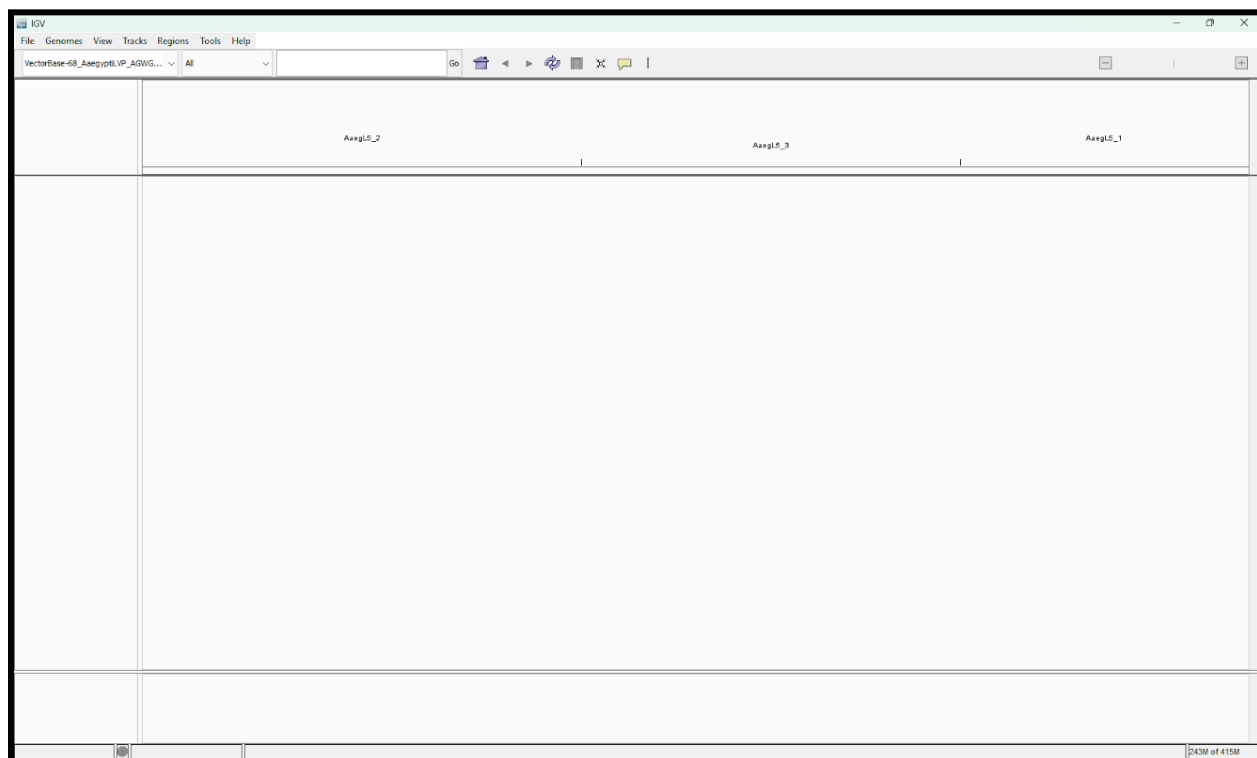
1. **Open IGV** by clicking on the shortcut/executable. (or running the shell script if you are on Linux) You should see a screen like this:



2. **Download the genome to the computer you're working on.** (You can likely load it from the NAS if you have no other option, but it will be fairly slow, since the genome file is around 1.3 gigabytes) The path to the current genome in the NAS is:
`NAS_Server_Shared/InformaticsResources/Aedes_aegypti/VectorBase-68_AaegyptiLVP_AGWG_Genome.fasta` You should also download the FASTA index
`(VectorBase-68_AaegyptiLVP_AGWG_Genome.fasta.fai)` and put it in the same folder you downloaded the genome to. If you do not have a .fai file for your genome, IGV will make one for you.
3. **Load the genome into IGV** by clicking `Genomes>Load Genome from File...` and selecting your .fasta file.

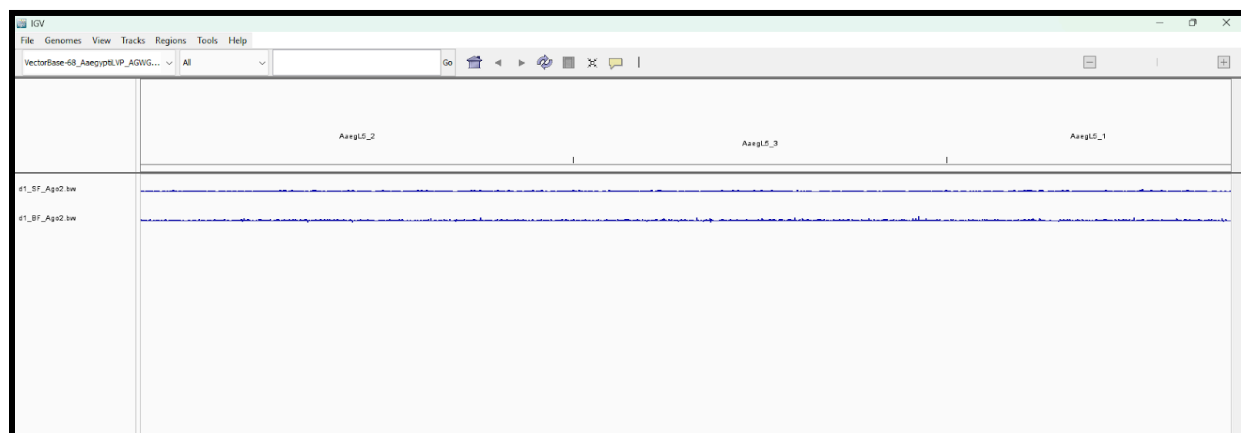


Your screen should look like this once it is finished loading:



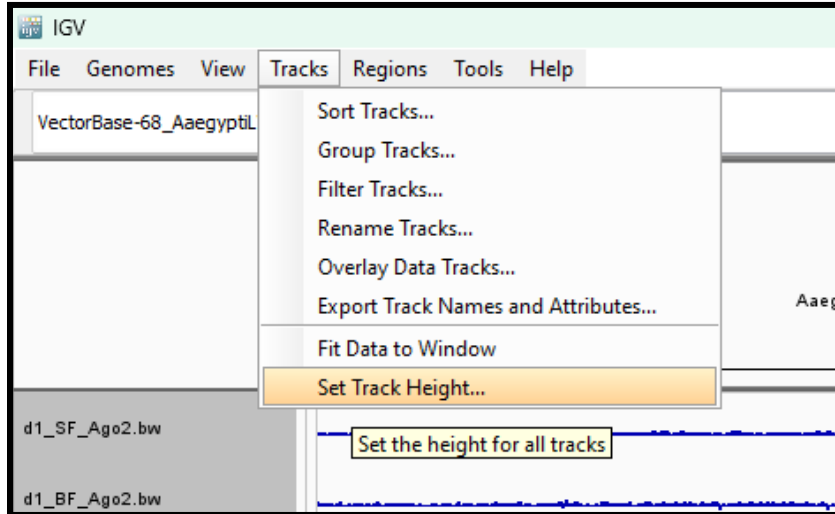
- 4. Load in your bigwig (.bw) files.** You can either download them locally or load them from the NAS. Drag/drop the files into either download bigwigs locally or drag/drop from the NAS. Drag/drop them into the largest rectangular region (where the blue signal lines are in the image below) or go to *File>Load from File...*

Your screen should now look like this:

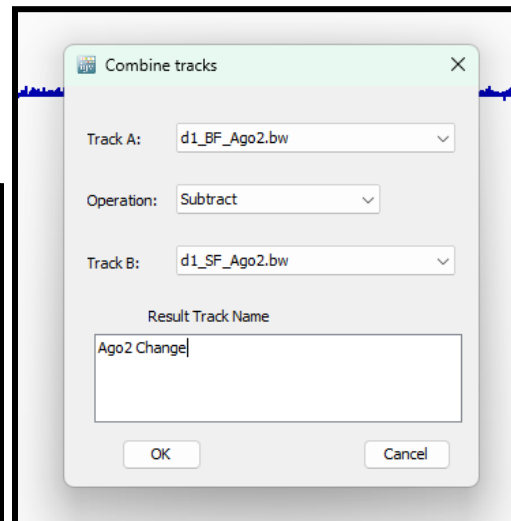
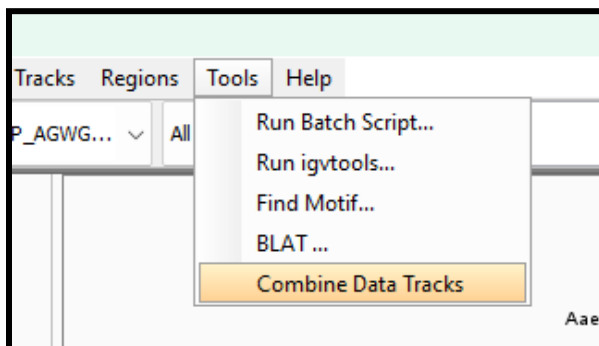


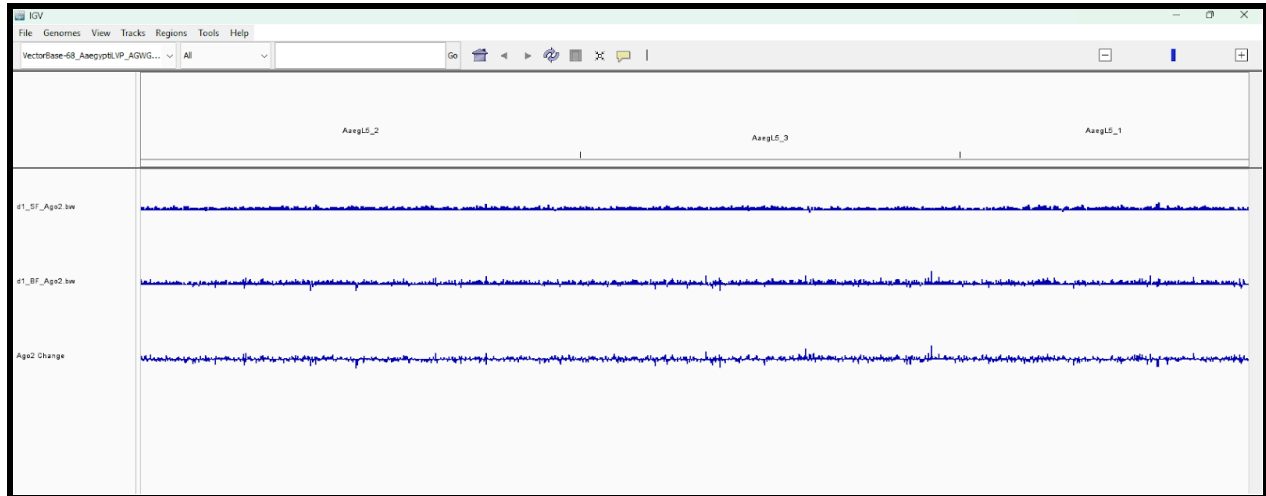
5. **Visualization Options:** These steps go over several options for changing the appearance of the IGV session and changing the way the data is visualized or visualizing new information. There are many, many more options than just these.

a. **Changing track height:** Go to *Tracks>Set Track Height...* (100 is a good default)



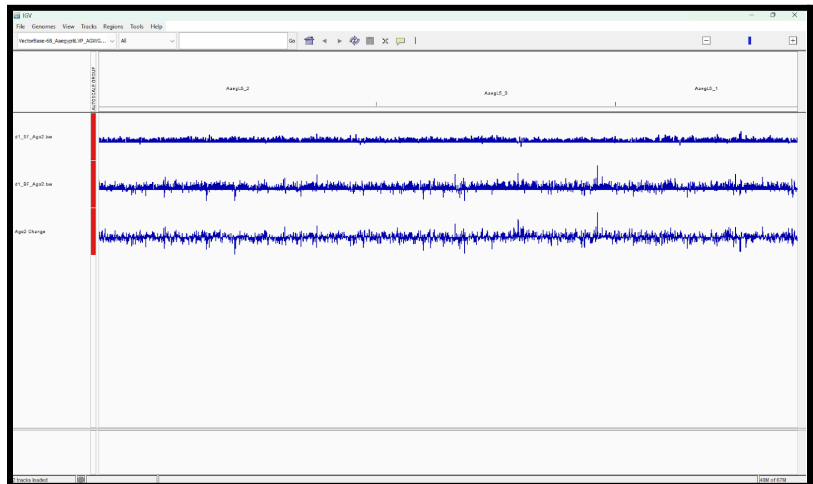
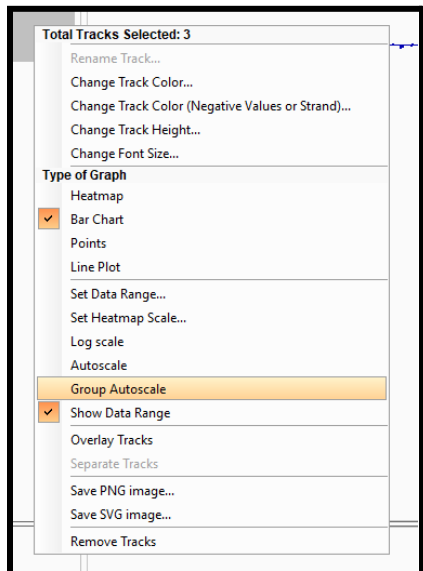
b. **Visualizing differences:** You can create a new data track that is the addition/subtraction/multiplication/division of two existing data tracks. To do this, go to *Tools > Combine Data Tracks*. Pick the desired tracks and operation. Track A is always the first term in the math expression. (A-B, A÷B, etc.)



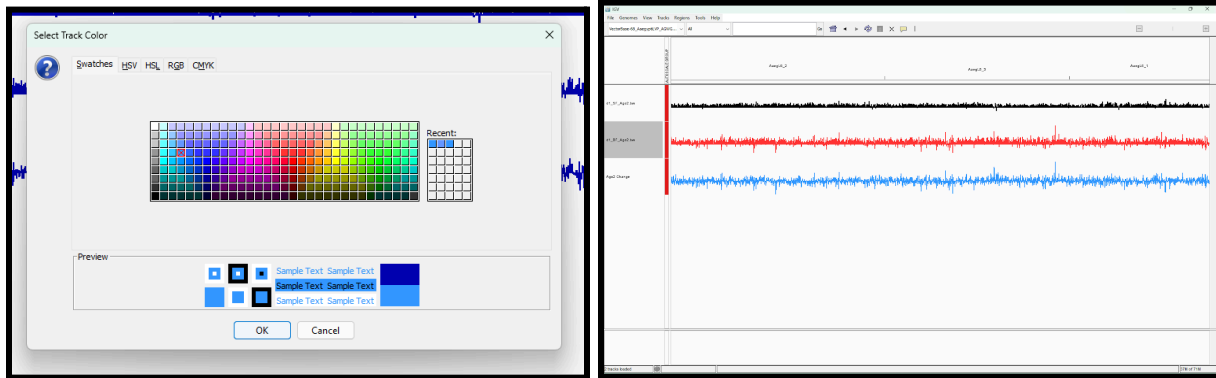


Your new, combined track should now appear. Be sure to visually inspect it to ensure the addition/subtraction/etc. is being done correctly.

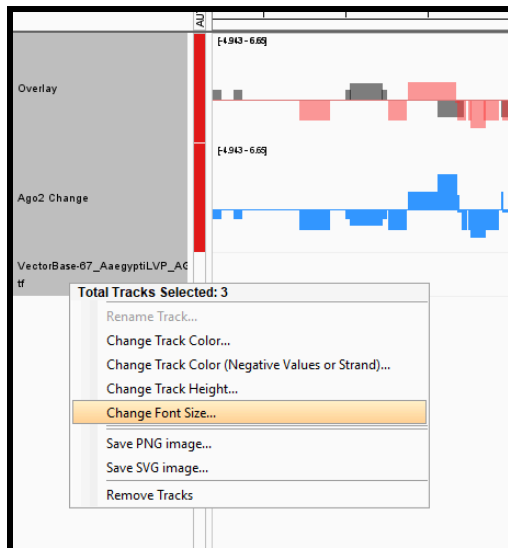
- c. Scaling tracks:** By clicking on one track and pressing shift or ctrl and clicking on another, you can select multiple tracks. Select all tracks and right click on one of the tracks' names, then click Group Autoscale.



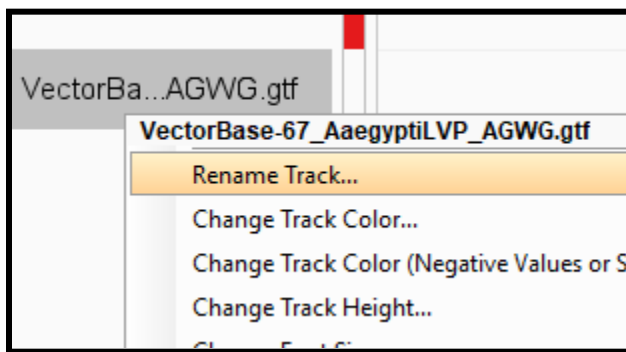
- d. **Changing track colors:** Right click a track's name, and click *Change Track Color*. The menu and results should look as follows:



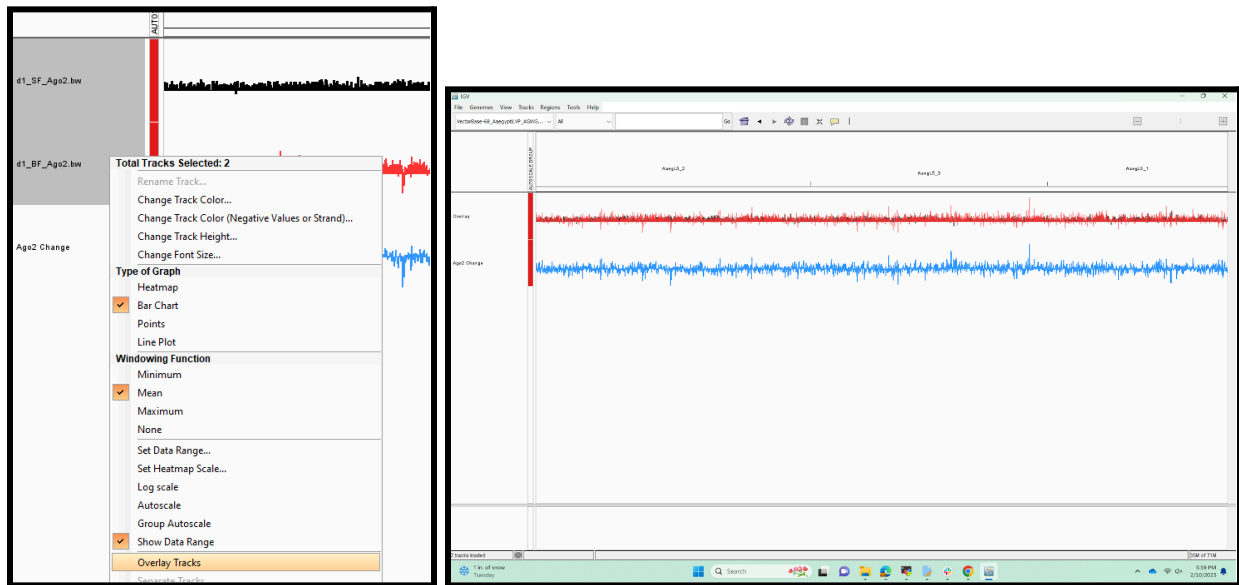
- e. **Scaling display text:** Select one or more tracks, right click their names, and click *Change Font Size...*



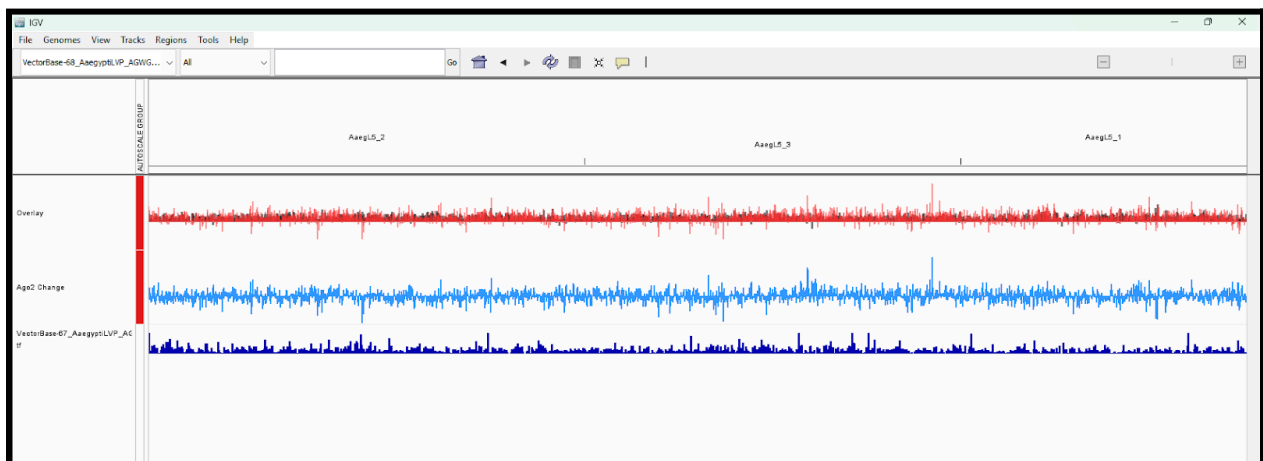
- f. **Renaming Tracks:** Right click a track name and click *Rename Track...*



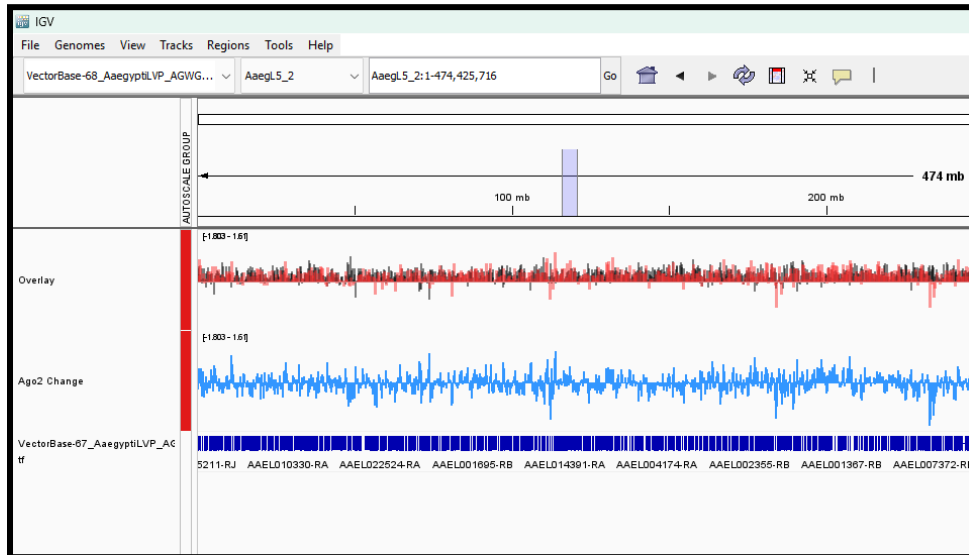
- g. Overlaying tracks:** Select two or more tracks to overlay with one another, right click their names and click **Overlay Tracks**. This will look a bit muddy in the genome-wide view, but it will look better after zooming in.



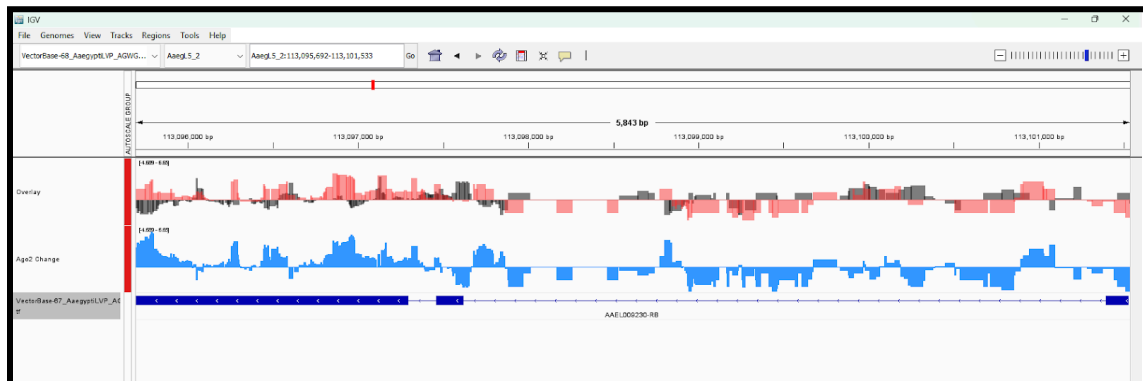
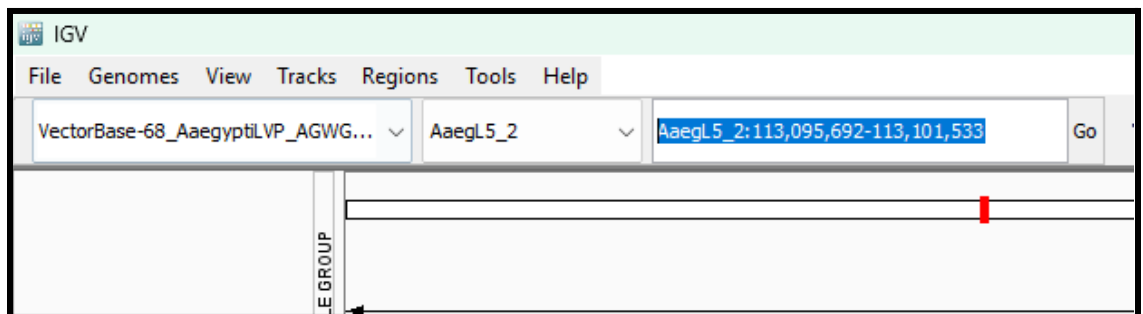
- h. Adding exon data.** By loading a .gtf,.bed, or .gff file in the same way you loaded bigwigs, you can show the locations, directions, and exons of genes. It will look a bit noisy at first, but will separate into different genes when we zoom in.



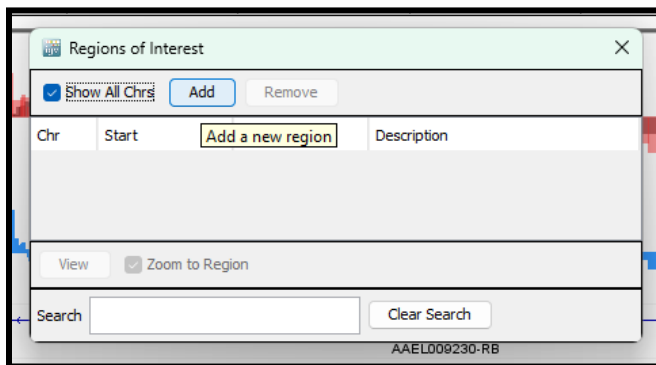
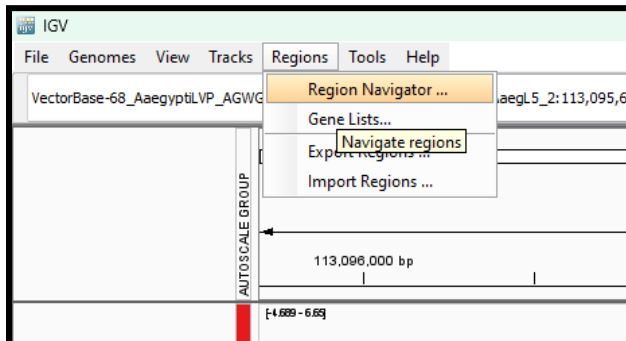
6. **Zooming/Navigating.** Click a chromosome of interest (above your tracks) to focus on it. You should now be able to click/drag across the top region to focus on different regions (unfortunately the cursor is not visible in these screenshots):



- a. **Manually enter the viewing frame ends.** Say we want to focus on the gene AAEL009230, which is on chromosome 2, starts at base 113,095,692, and ends on base 113,101,533. Type in your desired starts/ends for the viewing box at the top, separated by a hyphen. Commas are not necessary. Click **Go**.

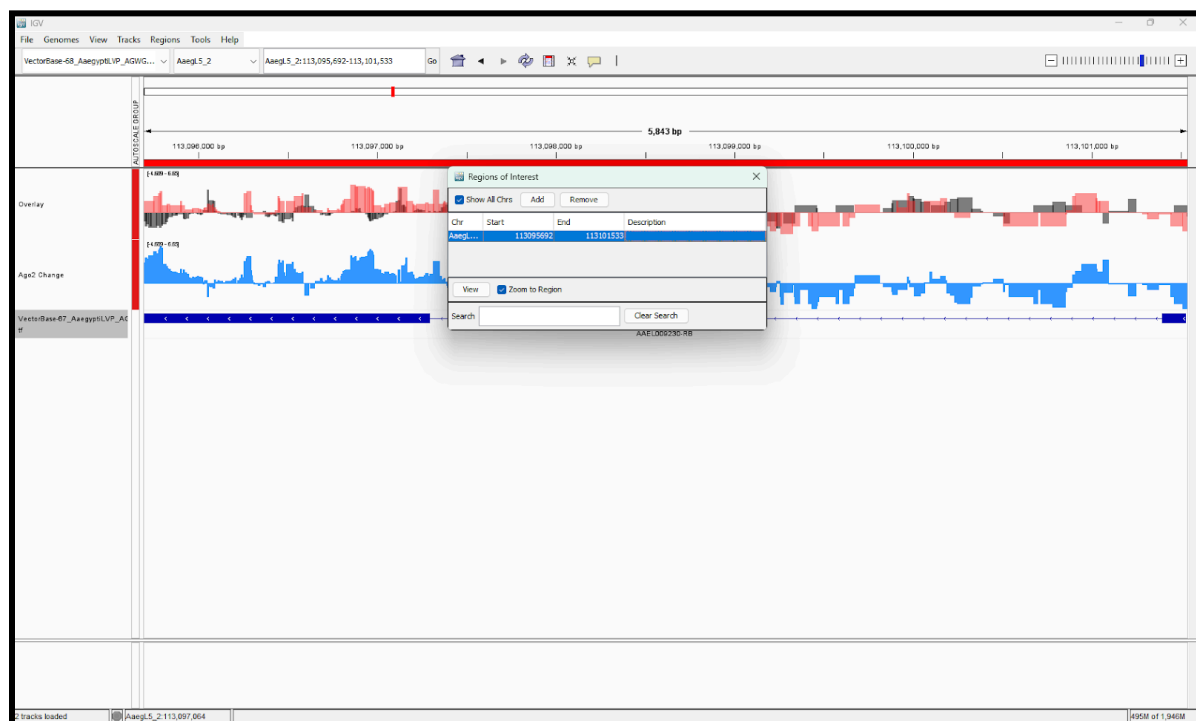


- b. Adding/Navigating Regions:** IGV regions allow you to bookmark and visually mark regions of interest. Go to **Regions>Region Navigator ...** to enter the region navigator, then click **Add** to add a new region. It will default to your current viewing frame, but you can manually edit it.

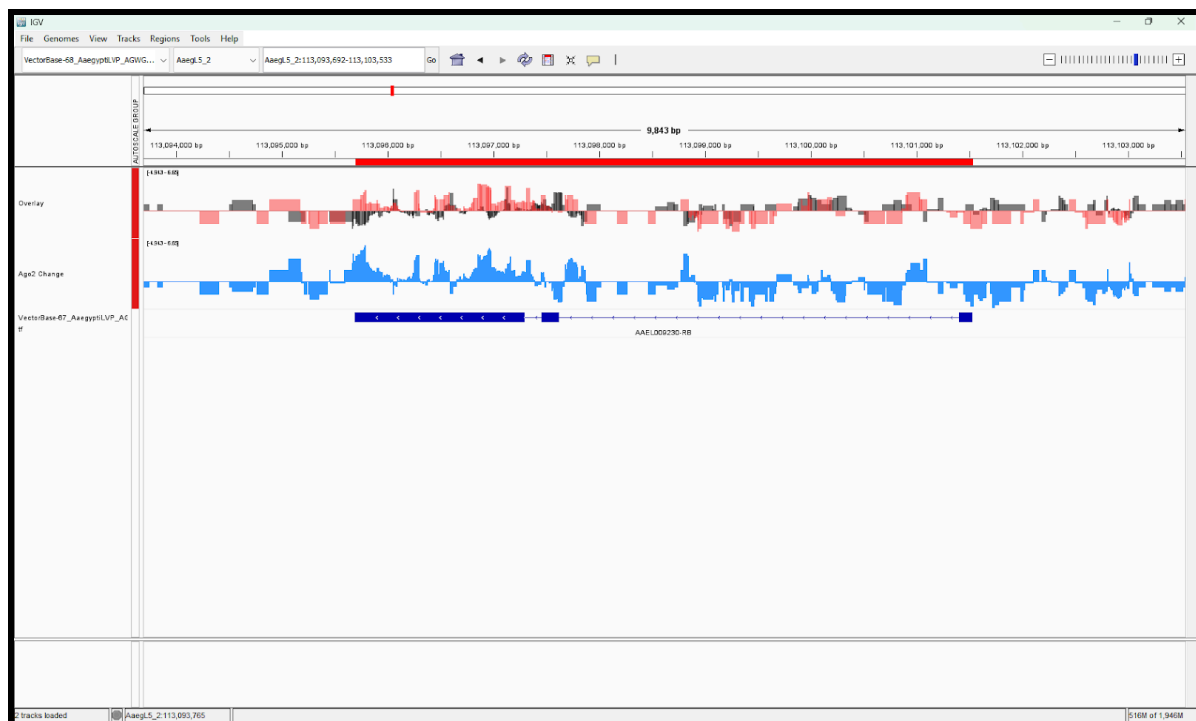


(Continued on next page...)

The new region will show up as a red line on top. You may have to click **View** for it to appear. This is useful for marking regions that aren't otherwise visually distinct.



Zooming out, we can see that this region spans the length of our gene. (Although this is redundant in this situation, since our GTF already shows this)



7. Saving visualizations or sessions:

- a. **Saving as a picture:** You can save PNGs or SVGs of your current visualization in the *File* menu as shown below. They will appear similar to a screenshot.



- b. **Saving your session:** **YOU CAN NOT SAVE WITH Ctrl+S. IGV WILL NOT WARN YOU THAT YOUR SESSION ISN'T SAVED BEFORE YOU CLOSE IT.** You must save by clicking *File>Save Session...* and saving as an XML file.

